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September 15, 1977
SUPERSEDING
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FEDERAL SPECIFICATION

CLEANING COMPOUND, OVEN

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. This specification covers three types of commercially available oven cleaners intended for degreasing and decarbonizing (see 6.1).

1.2 Classification. The oven cleaners shall be of the following types as specified (see 6.2).

Type 1 - Daytime (aerosol).
Type 2 - Overnight (aerosol).
Type 3 - Oven-grill cleaner (liquid).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

16 CFR 1500 - Administrative and Enforcement of Regulations for Hazardous Substances.
49 CFR 171-179 - Hazardous Materials Regulations.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 3069 - Delivery Rate of Aerosol Products.
D 3074 - Pressure in Metal Aerosol Containers.
D 3097 - Aerosol Products for Delivered Mass.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Types 1 and 2. The manufacturer is permitted discretion in formulating types 1 and 2 aerosol oven cleaners provided the products are in accordance with commercial practice. Type 1 oven cleaner (daytime) and type 2 (overnight) shall meet the specified requirements of section 3.

3.1.1.1 Propellant. The propellant shall be a hydrocarbon only. The use of vinyl chloride or fluorocarbons as the aerosol propellant gas are prohibited. The propellant shall be analyzed for fluorocarbons and vinyl chloride as specified in 4.3.7.

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3.1.1.2 Dispenser. The contents shall not react with nor be reacted upon by the interior surface of the dispenser or any of the dispenser components.

3.1.1.2.1 Container. The container shall be a commercial-type metal aerosol container of such construction as to assure acceptance of the finished package by common carriers operating in interstate commerce. The capacity of the container shall be sufficient to contain the specified net contents of the product with outage in accordance with commercial practice. The container shall not be overcoated to obliterate the original label.

3.1.1.2.2 Dispensing valve. The container shall have a continuous-spray dispensing valve with the performance characteristics specified in table I. The valve stem shall be fitted with a side discharge plastic actuator. The direction of the discharge shall be indicated either by the shape of the actuator or the mark on it. The actuator orifice shall be round and free of all obstructions which will affect the flow and discharge pattern of the contents (see 4.3.1). The dispensing valve shall be designed so that, when activated, none of the aerosol spray shall impinge on any part of the container. On operating the dispensing valve, the contents shall be ejected only through the orifice (see 4.3.1).

TABLE I. Valve performance characteristics

Characteristic	Requirement
Delivery rate at 21°C (70°F)	1.0 to 3.5 grams per second (g.p.s.).
Leakage in 48 hours, maximum	3.0 milliliters (ml).

3.1.1.2.3 Cover cap. The dispensing valve shall be protected from accidental functioning and damage by a metal or plastic cap which shall be removable. The cap shall provide a clearance of not less than 1/8 inch (0.32 cm) above the dispensing valve (see 4.2.2.1).

3.1.1.3 Unit container pressure stability. The filled unit containers shall show no evidence of leakage, distortion, or other pressure related defects when placed in a water bath at 55° ± 1°C (131° ± 2°F) for a period of 1 hour (see 4.3.3).

3.1.1.4 Net contents (Section 5, level A, only). The pressurized container shall contain 8 ± 0.3 oz. avoirdupois (217.8 ± 8.7g) or 16 ± 0.64 oz. avoirdupois (435.6 ± 17.4 g), as specified (see 6.2), of dispensible contents at the time of first acceptance by Government when tested as specified in 4.3.4.

3.1.2 Type 3. The oven cleaner shall be a uniform alkaline liquid, free of sediments and suspended particles. It shall consist of synthetic detergents, caustic and water-miscible solvents and meet the requirements as specified in 3.2, 3.3.2, and 3.4.

3.1.2.1 Unless otherwise specified, the supplier shall furnish a commercial-type hand Trigger Spray Unit in each shipping container to dispense the product from a one-gallon bottle.

3.2 pH value (all types). The pH values for the oven cleaners shall be not less than 13.0 when tested as specified in 4.3.5.

3.3 Cleaning performance.

3.3.1 Types 1 and 2. The cleaning performances for type 1 oven cleaner and type 2 oven cleaner shall be not less than 70 percent and 95 percent respectively when tested as specified in 4.3.6.

3.3.2 Type 3. The cleaning performance for the liquid oven cleaner shall be not less than 70 percent when tested as specified in 4.3.6.

3.4 Laws and regulations (all types). The product and container shall conform to the Federal Hazardous Substances Act and Department of Transportation Hazardous Materials Regulations. Each container of product of specified type shall be durably and legibly marked with precautions and directions for use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Quality conformance inspection.

4.2.1 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.2.2 Examination and test of the end item. The end item shall be examined in accordance with the classification of defects, inspection levels, and acceptable quality levels (AQL), set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled primary containers of oven cleaner of the same type contained herein.

4.2.2.1 Examination of the end item unit container for defects (types 1 and 2). The sample unit shall be one filled unit container. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level I, AQL 2.5 expressed in terms of percent defective.

Examine	Defects
Unit containers	Corroded, bent, distorted, or leaking. Not useable. Not as specified.
Valve	Not as specified.
Plastic actuator	Not as specified.
Cover cap	Not as specified.

4.2.2.2 Examination of the end item for defects in appearance of oven cleaner (type 3). The sample unit for this examination shall be one open filled primary container. The inspection level shall be S-2 with an AQL of 2.5 expressed in terms of percent defective.

Examine	Defects
Appearance of oven cleaner	Not a liquid. Presence of dirt or foreign matter. Not homogeneous. Any evidence of segregation. Not clear. Not pourable, too viscous.

4.2.2.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5. Defects shall be scored as specified in table II. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one shipping container fully prepared for delivery. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

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TABLE II. Examination of preparation for delivery

Examine	Defects
Container	Not as specified.
Content	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location sequence, or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper, or missing. Inadequate closure.

4.2.2.4 Examination for closure, waterproofing, and banding of shipping container. When shipping containers are required to comply with PPP-B-636, examination for defects in closure, water-proofing, and banding shall be made in accordance with the appendix to that specification.

4.2.2.5 Testing the end item. The end item shall be tested for the characteristics listed in table III for each lot presented for examination for each type of material. Two unit containers of the oven cleaners of type 1 or type 2 shall be drawn from not less than six separate shipping containers, selected throughout the lot presented for test. The test sample shall consist of not less than 12 unit containers. One pint of oven cleaner of type 3 shall be drawn from not less than five separate shipping containers, selected at random throughout the lot presented for test and composited. The test sample shall consist of a 4-pint composite of type 3 oven cleaner collected in a glass bottle.

TABLE III. Instructions for testing the end item

Characteristics	Specification reference: Requirement	Specification reference: Test method	Rqmts. appl. to: Indiv. unit	Rqmts. appl. to: Lot average	Results reported as: Pass or fail 1/	Results reported as: Numerically to nearest 2/
Delivery rate	Table I	4.3.1	X			0.5 ml
Aerosol spray impingement on container	3.1.1.2.2	4.3.1	X		X	
Discharge of material other than through orifice	3.1.1.2.2	4.3.1	X		X	
Leakage	Table I	4.3.2	X		X	
Filled unit container pressure stability	3.1.1.3	4.3.3	X		X	
Net weight of dispensible contents	3.1.1.4	4.3.4	X			0.1 gram
pH value	3.2	4.3.5		X		0.1 pH unit
Cleaning performance						
Types 1 and 2	3.3.1	4.3.6	X			1.0 percent
Type 3	3.3.2	4.3.6	X			1.0 percent
Propellant for fluorocarbon and vinyl chloride	3.1.1.1	4.3.7	X		X	

1/ If failure is indicated, report either description of failure or numerical point of failure, as applicable.
2/ Test report shall include all values on which results are based.

4.2.2.5.1 Unless otherwise specified, duplicate determinations shall be performed on the composite of type 3 oven cleaner with no evidence of failure to meet the specified requirements.

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4.3 Test procedures.

4.3.1 Delivery rate. Determine the delivery rate of the filled aerosol container in accordance with ASTM D 3069. Examine the container for aerosol spray impingement on any part of the container and discharge of material other than through the actuator orifice.

4.3.2 Leakage. Remove the cover cap from the unit container. Submerge the container in a water bath that is deep enough to allow the container to stand upright and be completely covered with water. Fill a wide-mouth graduated cylinder (graduated in milliliters) with water, invert it, and place it over the container so that it does not touch the valve and actuator. Maintain the temperature of the water bath at 21°C (70°F). Allow the container and graduate to stand for 48 hours.

4.3.3 Unit container pressure stability. Submerge filled unit container in an automatically controlled, circulating, constant temperature bath at 23° ± 3°C (73.4° ± 6°F). Place a safety shield in front of the bath in case the container ruptures. Set the bath at 55° ± 1°C (131° ± 2°F) and allow the temperature to rise. Place the containers in the bath for a period of 1 hour. Using gloves and face shield, remove the container from the bath and examine it for leakage, distortion, or other pressure related defects.

4.3.4 Net weight of dispensible contents. Determine the net weight of not less than two unit containers of high- or low-viscosity product in accordance with ASTM D 3097.

4.3.5 pH value. Determine the pH of the oven cleaners with a pH meter, using standard buffer solutions of pH 10 and pH 7 to standardize the pH meter.

4.3.6 Cleaning performance.

4.3.6.1 Soil formulation.

Pure ground beef (75% lean, 25% fat)	120 g
Grated cheddar cheese (Kraft or equal)	120 cc
Whole milk, homogenized	120 cc
White sugar, granulated	120 cc
Cherry juice (canned sour pie cherries)	120 cc
Instant tapioca, dry, uncooked	30 cc
Raw egg (without shell)	1
All-purpose flour	30 cc
Tomato juice	120 cc

The above ingredients are mixed together using a regular hand food mixer on low speed. The mixture is well mixed, but still has a lumpy consistency. Prepared soil or panels shall not be used after 5 days of their preparation.

4.3.6.2 Preparation of soiled panels. White porcelain test panels, 4.5 x 4.5 inches (11.2 x 11.2 cm) may be obtained from Erie Ceramics Art Company, 3120 West 22nd Street, Erie, PA 16505 or Challenge Stamping and Porcelain Company, Grand Haven, MI 49417.

4.3.6.2.1 Rate of application. The rate of application for the soil shall be 0.78 g/panel using a 2-inch (5.2-cm) nylon brush for soil application.

4.3.6.2.2 Panel aging. The soiled panels shall be baked uncovered in a controlled temperature oven at 232.2°C (450°F) for one hour, then removed from the oven and cooled to room temperature.

4.3.6.3 Cleaning performance. Nine soiled panels, as specified in 4.3.6.2.2, shall be used for each test in accordance with the manufacturer's use directions except that type 3 oven cleaner shall be tested only under warm electric oven conditions. The soiled panels are placed in a horizontal position inside the oven and conditioned according to the use directions. Then, the soiled panels are sprayed with 3.2 g of the oven cleaner and maintained at the temperature according to directions. Damp sponge lightly. Percent soil removal is estimated visually in relation to the surface cleaned. The nine panels are then averaged.

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4.3.7 Analysis of propellant gas for fluorocarbons and vinyl chloride.

4.3.7.1 Special apparatus and materials. The chemical analysis procedure described herein requires the following apparatus and materials:

A can piercing, pressure measuring device for determining the internal pressure of aerosol products. The device is available as Model CBA-6 from Builders Sheet Metal Works, Inc., 110 Wooster Street, New York, New York 10012.

A temperature programmable gas chromatograph (Hewlett-Packard Model 5750 gas chromatograph, or equivalent) equipped with a 0.5 ml gas sampling valve and thermal conductivity detector.

Two stainless steel gas chromatography columns, 6 feet long by 1/8" outside diameter, packed with 60-200 mesh silica gel.

Standard propellant gases in lecture bottles from Matheson Gas Products (or another supplier): vinyl chloride, fluorocarbon Nos. 11, 12, 114, 115.

Matheson lecture bottle control valve Model 58, or equivalent.

4.3.7.2 Gas chromatographic analysis. Condition the chromatographic columns at 200°C with normal helium as the carrier gas at 75-100 ml/minute for one hour. Remove the actuator and cap of an unused aerosol container. Puncture the can using the pressure measuring device and connect the device to the gas chromatograph. Triplicate chromatographic analyses shall be done on the aerosol propellant sample and on mixtures of the aerosol propellant with each of the five standard propellant gases. For mixing the standard propellants with the aerosol propellant sample, use a "T" connection (Swagelock or equivalent). The "T" connection shall be removed entirely when the aerosol propellant sample alone is to be analyzed. Bubble each gas sample through the sampling valve at a moderate rate for two minutes before each run is made to purge the valve of any other gases. Analyze the samples using a thermal/conductivity detector. The temperature program shall be 100°C and a 5-minute upper limit hold at 200°C. Helium is used as the carrier gas at the rate of 75-100 ml per minute. There shall be no detectable material in the aerosol propellant sample having the same retention time as any of the five standard propellant gases.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Types 1 and 2. The quantity of oven cleaner specified in 3.1.1.4 furnished in a container specified in 3.1.1.2 does not require additional packaging.

5.1.1.2 Type 3. One gallon of oven cleaner shall be packaged in a high density (min. 0.95) polyethylene bottle with a handle and a closure with a protective recloseable feature. Each bottle shall be secured with a seal A closure conforming to PPP-C-186.

5.1.2 Commercial. The oven cleaner shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling, and storage.

5.2 Packing. Packing shall be level A or Commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Types 1 and 2. Twelve unit containers of oven cleaner of one type only, packaged as specified in 5.1.1.1, shall be packed in a box conforming to PPP-B-636, class weather resistant. The box shall be fitted with dividers, partitions, or separators fabricated of fiberboard of the same description as the box. The box shall be 1/8 to 3/8 inch higher than the containers. The box shall be closed, waterproofed, and banded in accordance with the appendix of that specification.

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5.2.1.2 Type 3. Four or six 1-gallon bottles of oven cleaner packaged as specified in 5.1.1.2 and one Trigger Spray Unit (see 3.1.2.1) shall be packed in a fiberboard shipping container conforming to PPP-B-636, style RSC-L, grade V2s. In addition, each container shall be provided with top and bottom pads fabricated from the same material as the box liner. The pads shall measure 1/4 inch less than the inside length and width of the box liner. Each shipping container shall be closed, waterproofed, and banded in accordance with the appendix to PPP-B-636.

5.2.2 Commercial. The packaged oven cleaner shall be packed in fiberboard boxes to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.3 Marking. Packages, shipping containers, and unitized loads (when applicable) shall be marked in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military agencies, as specified (see 6.2).

5.4 Unitization. When shipments to government depots are full car or truckload, the shipping container shall be unitized for shipment in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

6. NOTES

6.1 Intended use. The oven cleaners covered by this specification are intended for removal of carbon deposits, baked-on grease and other surface contaminants from cooking ovens and grills. Type 3 is U.S.D.A acceptable in meat and poultry plants, provided it meets U.S.D.A requirements.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required.
(c) Packaging and packing required (see 5.1 and 5.2).
(d) Marking required (see 5.3).
(e) Weight of contents per unit container (see 3.1.1.4).

MILITARY INTEREST:

DOD has waived coordination of this document until further notice.

Preparing Activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

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